

Performance Testing

Date	4 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Streamlit Application (Manual Performance Testing)
Type of Testing	Load Testing (Functional Performance Evaluation)
Target Module / API	Audio Upload, Speech-to-Text, Semantic Analysis, Waveform Generation, PDF Report Generation
Test Environment	Local Machine (Windows 10, Python 3.x, Streamlit)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Upload audio and perform speech transcription	1	20	Audio successfully transcribed
2	Generate semantic similarity score	1	15	Concept score displayed
3	Generate waveform visualization	1	5	Waveform displayed successfully
4	Generate downloadable PDF report	1	10	PDF created without errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.4 sec	Pass	Fast response for UI operations
2	Response Time (Max)	< 5 seconds	3.2 sec	Pass	Report generation completed successfully
3	Throughput (Req/sec)	1 Req/sec	1.2 Req/sec	Pass	Stable for single-user testing
4	Error Rate	< 1%	0%	Pass	No runtime errors observed
5	CPU Utilization	< 80%	58%	Pass	CPU usage remained within limits
6	Memory Utilization	< 80%	62%	Pass	Memory consumption remained stable

Step 4: Observations & Analysis

- The application successfully processed uploaded audio files without noticeable delays.
- Speech transcription and semantic similarity evaluation were completed correctly.
- Waveform visualization was generated quickly and displayed accurately.
- PDF reports were created successfully and saved to the reports directory.
- No application crashes or runtime errors occurred during testing.
- CPU and memory utilization remained within acceptable limits, indicating efficient resource usage.
- The application is suitable for educational use with single or light concurrent users.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

localhost:8501

Voice-Based Communication Understanding Analyzer

Automated evaluation of spoken conceptual explanations using AI.

Upload Student Audio (WAV/MP3)

Upload 200MB per file - WAV, MP3

Concept Reference

Analyze Concept Understanding

Transcribed Explanation

The stale smell of old beer lingers. It takes heat to bring out the odor. A cold dip restores health and zest. A salt pickle tastes fine with ham. Tacos al pastor are my favorite. A zestful food is the hot cross bun.

Final Result

5.45/100

Poor Understanding

Grade : Poor

Semantic Similarity	Duration (sec)	RMS Energy
0.05	18.36	0.0327

Audio Waveform